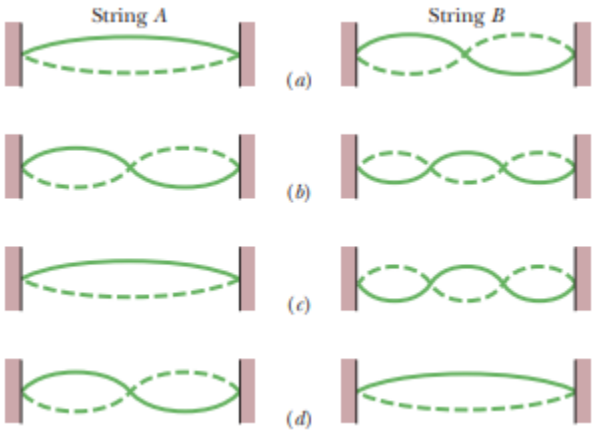
1.Strings A and B have identical lengths and linear densities, but string B is under greater tension than string A. Figure shows four situations, (a) through (d), in which standing wave patterns exist on the two strings. In which situations is there the possibility that strings A and B are oscillating at the same resonant frequency?



2. A generator at one end of a very long string creates a wave given by

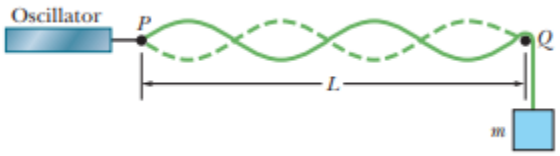
$$y = (6.0\text{ cm})\cos\frac{\pi}{2}[(2.00\text{ m}^{-1})x + (8.00\text{ s}^{-1})t],$$

and a generator at the other end creates the wave

$$y = (6.0\text{ cm})\cos\frac{\pi}{2}[(2.00\text{ m}^{-1})x - (8.00\text{ s}^{-1})t].$$

Calculate the (a) frequency, (b) wavelength, and (c) speed of each wave. For x 0, what is the location of the node having the (d) smallest, (e) second smallest, and (f) third smallest value of x? For x ≥ 0, what is the location of the antinode having the (g) smallest, (h) second smallest, and (i) third smallest value of x?

2. In Fig. a string, tied to a sinusoidal oscillator at P and running over a support at Q, is stretched by a block of mass m. Separation L =1.20 m, linear density m = 1.6 g/m, and the oscillator

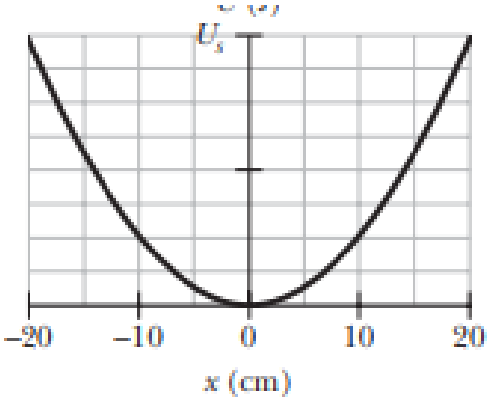


frequency f = 120 Hz. The amplitude of the motion at P is small enough for that point to be considered a node. A node also exists at Q. (a) What mass m allows the oscillator to set up the fourth harmonic on the string? (b) What standing wave mode, if any, can be set up if m = 1.00 kg?

3. An automobile can be mounted on four identical springs as far as vertical oscillations are concerned. The springs of a certain car are adjusted so that the oscillations have a frequency of 3.00 Hz. (a) What is the spring constant of each spring if the mass of the car is 1450 kg and the mass is evenly distributed over the springs? (b) What will be the oscillation frequency if five passengers, averaging 73.0 kg each, ride in the car with an even distribution of mass?

4. A block rides on a piston (a squat cylindrical piece) that is moving vertically with simple harmonic motion. (a) If the SHM has period 1.0 s, at what amplitude of motion will the block and piston separate? (b) If the piston has an amplitude of 5.0 cm, what is the maximum frequency for which the block and piston will be in contact continuously?

5. Figure 15-38 gives the one-dimensional potential energy well for a 2.0 kg particle (the function U(x) has the form bx^2 and the vertical axis scale is set by Us = 2.0 J). (a) If the particle passes through the equilibrium position with a velocity of 85 cm/s, will it be turned back before it reaches x = 15 cm? (b) If yes, at what position, and if no, what is the speed of the particle at x = 15 cm?



6. At a certain harbor, the tides cause the ocean surface to rise and fall a distance d (from highest level to lowest level) in simple harmonic motion, with a period of 12.5 h. How long does it take for the water to fall a distance 0.250d from its highest level?

7. A block rides on a piston (a squat cylindrical piece) that is moving vertically with simple harmonic motion. (a) If the SHM has period 1.0 s, at what amplitude of motion will the block and piston separate? (b) If the piston has an amplitude of 5.0 cm, what is the maximum frequency for which the block and piston will be in contact continuously?

8. Two particles oscillate in simple harmonic motion along a common straight-line segment of length A. Each particle has a period of 1.5 s, but they differ in phase by π/6 rad. (a) How far apart are they (in terms of A) 0.50 s after the lagging particle leaves one end of the path? (b) Are they then moving in the same direction, toward each other, or away from each other?

9. Two particles execute simple harmonic motion of the same amplitude and frequency along close parallel lines. They pass each other moving in opposite directions each time their displacement is half their amplitude. What is their phase difference?

10. A block of mass M = 5.4 kg, at rest on a horizontal frictionless table, is attached to a rigid support by a spring of constant k = 6000 N/m. A bullet of mass m = 9.5 g and velocity of magnitude 630 m/s strikes and is embedded in the block (Fig below). Assuming the compression of the spring is negligible until the bullet is embedded, determine (a) the speed of the block immediately after the collision and (b) the amplitude of the resulting simple harmonic motion.

